

Namal University Transport Management System (NUTMS)

Database Design Document

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION :

Namal University, Mianwali, is a private sector university that provides daily transport services to its students, faculty, and staff for commutes between the main campus, hostels, and nearby city areas. Currently, all transport operations are managed through manual, paper-based processes including route planning, driver assignments, trip scheduling, and reporting which introduce errors, delays, data inconsistency, and lack of transparency across the system.

The Namal University Transport Management System (NUTMS) is a web-based platform designed to digitize and automate these transport operations. This Database Design Document focuses on the database design layer of NUTMS the backbone responsible for storing, managing, and reliably retrieving all transport-related data.

The system targets four distinct user classes: Students, who track buses and receive schedule notifications; Faculty, who additionally request special vehicle bookings; Drivers, who view assigned routes and update trip statuses; and Transport Administrators, who manage all users, vehicles, routes, and schedules through a centralized dashboard.

The database will be implemented using MySQL a relational database management system offering ACID compliance, advanced indexing, and robust integrity constraints — supported by a Node.js/Express.js backend, a React.js frontend, OpenStreetMap for geospatial visualization, and Socket.io for real-time GPS communication. Cloud hosting on Vercel ensures high availability throughout academic semesters.

1.2. PROBLEM STATEMENT :

Namal University currently manages its transport operations through manual, paper-based processes without any centralized database system, resulting in inefficient and unreliable data management. Information related to routes, schedules, drivers, and vehicles is maintained in physical registers and informal spreadsheets, leading to data inconsistency, duplication, and lack of a single authoritative source. Furthermore, there is no real-time visibility for students and faculty to track bus locations, schedule updates, or delays, as communication relies heavily on informal and slow methods such as phone calls. The system also imposes a high administrative burden, requiring significant manual effort for report generation and resource allocation, thereby increasing the likelihood of human errors. Additionally, the absence of audit trails and accountability mechanisms makes it difficult to track user actions, booking histories, or resolve disputes effectively. As the university grows, the current manual system lacks scalability to handle increasing users, vehicles, and route complexities. Therefore, there is a critical need for a structured, secure, and role-based MySQL relational database system (NUTMS) that ensures centralized data management, consistency, real-time accessibility, and operational efficiency.

1.3. PROJECT OBJECTIVES :

General Objective : To design and implement a centralized, relational database system for the Namal University Transport Management System (NUTMS) that automates and digitizes all transport operations, enabling reliable data storage, real-time information access, and data-driven decision making for all stakeholders.

Specific Objectives:

1. **Real-Time GPS Data Storage:** Design a GPS_Log table capable of storing location updates every 5-10 seconds per bus, with indexing on bus ID and timestamp so that the latest bus location can be retrieved within 5 seconds.
2. **Role-Based Access Control :** Define separate access levels in the database for Students, Faculty, Drivers, and Administrators so that each user can only view and perform operations that their role permits.
3. **Route and Schedule Management:** Enable administrators to create, update, and manage bus routes, stop sequences, departure times, and arrival times through structured relational tables with referential integrity.
4. **Driver and Vehicle Record Maintenance:** Maintain complete records of all drivers and vehicles, including vehicle capacity, registration details, license information, and driver-to-route assignments.
5. **Special Vehicle Booking for faculty :** Support a faculty booking request pipeline — from submission and pending status through administrative approval or rejection — with full notification triggers.
6. **Comprehensive Audit Logging:** Record all critical user actions (bookings, cancellations, approvals, login attempts, profile updates) with user ID, action type, and timestamp to ensure full accountability and transparency.
7. **Report Generation Support :** Enable the database to efficiently run analytical queries that generate daily, weekly, and monthly reports on transport usage, peak hours, fuel consumption, and driver trip completion rates.
8. **Data Availability and Security:** Maintain 99% database availability during academic semesters through cloud hosting, implement daily automated backups with 30-day retention, and encrypt all sensitive user data.

GITHUB REPOSITORY LINK:

<https://github.com/Altaf-Hussain-11/DB-Project-NUTMS-.git>

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY:

Sr. No	Entity Name	Description
01	User	A USER is a person who has a registered account in NUTMS. Every user has a unique role (Student, Faculty, Driver, or Administrator) and a set of access permissions determined by that role. A USER is identified by a unique User_ID.
02	STUDENT	A STUDENT is a subtype of USER who is enrolled at Namal University and uses NUTMS primarily to track buses and receive schedule notifications. A STUDENT inherits USER attributes and additionally stores enrollment status and residential type.
03	FACULTY	A FACULTY is a subtype of USER who is an academic or administrative staff member. In addition to bus tracking, a FACULTY member can submit special vehicle booking requests. A FACULTY inherits USER attributes and additionally stores employee ID and department.
04	DRIVER	A DRIVER is a subtype of USER who operates university buses. A DRIVER views assigned routes, updates trip statuses, and provides real-time information. A DRIVER inherits USER attributes and additionally stores license number and expiry date.
05	ADMINISTRATOR	An ADMINISTRATOR is a subtype of USER with full system access. Administrators manage users, routes, schedules, vehicles, and generate operational reports. An ADMINISTRATOR inherits USER attributes and additionally stores admin level.
06	VEHICLE	A VEHICLE is a physical vehicle owned or operated by Namal University for transport services. A VEHICLE is identified by a unique Vehicle_ID and stores registration plate, seating capacity, model, and current operational status.
07	ROUTE	A ROUTE defines a named path taken by university buses between a set of stops. A ROUTE is identified by a Route_ID and includes start/end locations, total distance, and estimated travel duration.
08	STOP	A STOP is a designated physical location on a ROUTE where passengers board or alight. A STOP is identified by a Stop_ID and stores the stop name, GPS coordinates, and sequence number within its route.
09	SCHEDULE	A SCHEDULE defines a specific departure time for a

		Vehicle operating on a ROUTE on given day(s) of the week. A SCHEDULE is identified by a Schedule_ID.
10	TRIP	A TRIP is a single, real-world execution of a SCHEDULE by a specific BUS and DRIVER on a specific date. A TRIP tracks status (Scheduled, In Progress, Completed, Cancelled), actual departure, and actual arrival.
11	GPS_LOG	A GPS_LOG is a time-stamped record of a bus's geographical position. The system stores location updates every 10–15 seconds for each active bus.
12	BOOKING	A BOOKING is a faculty member's formal request for a special vehicle for a trip outside the regular schedule. A BOOKING records destination, trip date, purpose, passenger count, and approval status.
13	NOTIFICATION	A NOTIFICATION is a system-generated or administrator-triggered message sent to one or more users. It stores message content, type (delay, schedule change, emergency, booking update), and delivery timestamp.
14	AUDIT_LOG	An AUDIT_LOG is a record of every critical action performed within the system. It captures the acting user, action type, affected entity, old value, new value, IP address, and timestamp.
15	VEHICLE_MAINTENANCE	VEHICLE_MAINTENANCE record captures every maintenance or service event performed on a university bus. It stores the type of maintenance, a description, the date it was carried out, the cost.
16	FUEL_CONSUMPTION	FUEL_CONSUMPTION record logs every fuel refill event for a university bus. It tracks the quantity of fuel filled, the cost, and the timestamp. This entity supports fuel usage analysis and route efficiency reporting

DATA DICTIONARY:

2.2.1. User

Sr.	Name	Data Type	Constraint	Description
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Sr.	Name	Data Type	Constraint	Description
01	User_ID (PK)	INT	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique system-generated identifier for each user.
02	Full_Name	STRING	NOT NULL	Full legal name of the user.
03	Email	VARCHAR(150)	NOT NULL, UNIQUE	Official Namal University email address used for login and notifications.
04	Password_Hash	VARCHAR(255)	NOT NULL	Bcrypt-hashed password; plain text is never stored.
05	Role_id	ENUM	NOT NULL	Role_id of the user: Student, Faculty, Driver, or Administrator.
06	Profile_Photo	VARCHAR(255)	OPTIONAL	File path to the user's profile picture.
07	Created_At	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp when the account was created.
08	Last_Login	DATETIME	OPTIONAL	Timestamp of the user's most recent successful login.

2.2.2. Student

Inherits all USER attributes. Additionally stores:

Sr.	Name	Data Type	Constraint	Description
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Sr.	Name	Data Type	Constraint	Description
01	Registration_No	VARCHAR(30)	NOT NULL, UNIQUE	Official university registration number (e.g., NUM-BSCS-2024-11).
02	Residential_Type	ENUM	NOT NULL	Either 'Hosteler' or 'Day Scholar'.

2.2.3. Faculty

Inherits all USER attributes. Additionally stores:

Sr.	Name	Data Type	Constraint	Description
01	Employee_ID	VARCHAR(20)	NOT NULL, UNIQUE	Official university employee identification number.
02	Department	STRING	NOT NULL	Academic or administrative department of the faculty member.
03	Designation	STRING	NOT NULL	Job title (e.g., Lecturer, Assistant Professor).
04	Office_Extension	VARCHAR(10)	OPTIONAL	Internal campus phone extension.

2.2.4. Driver

Inherits all USER attributes. Additionally stores:

Sr.	Name	Data Type	Constraint	Description
01	License_Number	VARCHAR(30)	NOT NULL, UNIQUE	Government-issued driver's license number.
02	License_Expiry	DATE	NOT NULL	Expiry date of the driving license.
03	Experience_Years	INT	NOT NULL, >= 3	Number of years of professional driving experience.
04	Employment_Status	ENUM	NOT NULL	Active, On Leave, or Terminated.

2.2.5. Administrator

Inherits all USER attributes. Additionally stores:

Sr.	Name	Data Type	Constraint	Description
01	Admin_Level	ENUM	NOT NULL	Super Admin or Regular Admin — controls override permissions.
02	Office_Location	VARCHAR(100)	OPTIONAL	Physical office location on campus.

2.2.6. VEHICLE

Sr.	Name	Data Type	Constraint	Description
01	Vehicle_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique system identifier for each bus.
02	Registration_Plate	VARCHAR(15)	NOT NULL, UNIQUE	Official vehicle registration plate number.
03	Make_Model	VARCHAR(80)	NOT NULL	Manufacturer and model of the bus (e.g., Hino 300).
04	Seating_Capacity	INT	NOT NULL, > 0	Maximum number of passengers the bus can carry.
05	Manufacturing_Year	YEAR	NOT NULL	Year the vehicle was manufactured.
06	Operational_Status	ENUM	NOT NULL	Active, Under Maintenance, or Retired.
07	GPS_Device_ID	VARCHAR(50)	OPTIONAL, UNIQUE	Identifier of the GPS hardware unit installed in the bus.

2.2.7. Route

Sr.	Name	Data Type	Constraint	Description
01	Route_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique system identifier for each route.
02	Route_Name	VARCHAR(100)	NOT NULL, UNIQUE	Descriptive name of the route (e.g., Campus–City Centre).
03	Start_Location	VARCHAR(150)	NOT NULL	Name/address of the starting point.
04	End_Location	VARCHAR(150)	NOT NULL	Name/address of the terminal point.
05	Total_Distance_km	DECIMAL(6,2)	NOT NULL, > 0	Total length of the route in kilometres.
06	Estimated_Duration_min	INT	NOT NULL, > 0	Estimated travel time from start to end in minutes.

2.2.8. Stop

Sr.	Name	Data Type	Constraint	Description
01	Stop_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique system identifier for each stop.
02	Stop_Name	VARCHAR(100)	NOT NULL	Common name for the bus stop or landmark.
03	Sequence_Number	INT	NOT NULL, > 0	Order of this stop within its route (1 = first stop).
04	Arrival_Offset_min	INT	NOT NULL, >= 0	Estimated Minutes from route departure time

Sr.	Name	Data Type	Constraint	Description
				when bus reaches this stop.

2.2.9. Schedule

Sr.	Name	Data Type	Constraint	Description
01	Schedule_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique system identifier for each schedule entry.
02	Departure_from	STRING	NOT NULL	The starting location from where the trip or route begins.
03	Destination	STRING	NOT NULL	The final location where the trip or route ends.
02	Departure_Time	TIME	NOT NULL	Scheduled departure time from the route's start location.
03	Days_Of_Week	STRING	NOT NULL	Days this schedule runs: Mon, Tue, Wed, Thu, Fri, Sat, Sun.
04	Effective_From	DATE	NOT NULL	Start date from which the schedule is valid.
05	Effective_Until	DATE	OPTIONAL	End date until which the schedule is valid (NULL = indefinite).

2.2.10. Trip

Sr.	Name	Data Type	Constraint	Description
01	Trip_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each trip instance.
02	Trip_Date	DATE	NOT NULL	Calendar date on which the trip is/was conducted.

Sr.	Name	Data Type	Constraint	Description
03	Trip_Status	ENUM	NOT NULL	Scheduled, In Progress, Completed, or Cancelled.
04	Departure_Time	DATETIME	OPTIONAL	Recorded actual departure time (set when trip starts).

2.2.11. GPS

Sr.	Name	Data Type	Constraint	Description
01	Log_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each GPS reading.
02	Timestamp	DATETIME	NOT NULL	Exact date and time the location reading was captured.
03	Latitude	DECIMAL(9,6)	NOT NULL	Latitude coordinate of the bus at time of log entry.
04	Longitude	DECIMAL(9,6)	NOT NULL	Longitude coordinate of the bus at time of log entry.
05	Speed_kmph	INT	OPTIONAL	Bus speed in kilometres per hour at time of reading.

2.2.12. Booking

Sr.	Name	Data Type	Constraint	Description
01	Booking_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each special vehicle booking request.
02	Destination	VARCHAR(200)	NOT NULL	Intended destination of the special trip.
03	Trip_Date	DATE	NOT NULL	Requested date for the special vehicle trip.
04	Departure_Time	TIME	NOT NULL	Requested departure time for the trip.
05	Purpose	TEXT	NOT NULL	Reason or justification for the special vehicle request.
06	Passenger_Count	INT	NOT NULL, > 0	Number of passengers expected on the trip.
07	Status	ENUM	NOT NULL	Pending, Approved, or Rejected.
08	Admin_Comments	TEXT	OPTIONAL	Administrator's remarks on approval or rejection.
09	Requested_At	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp when the booking request was submitted.
10	Resolved_At	DATETIME	OPTIONAL	Timestamp when the request was approved or rejected.

2.2.13. Notification

Sr.	Name	Data Type	Constraint	Description
01	Notification_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each notification.
02	Type	ENUM	NOT NULL	Delay, Schedule Change, Emergency, or Booking Update.
03	Title	VARCHAR(150)	NOT NULL	Short subject line of the notification.
04	Message	TEXT	NOT NULL	Full body content of the notification.
05	Is_Broadcast	BOOLEAN	NOT NULL	If TRUE, sent to all users; if FALSE, targeted to specific users.
06	Created_At	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp when the notification was created.
07	Sent_At	DATETIME	OPTIONAL	Timestamp when the notification was successfully dispatched.

2.2.14. Audit Log

Sr.	Name	Data Type	Constraint	Description
01	Log_ID (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each audit log entry.
02	Action_Type	ENUM	NOT NULL	e.g., LOGIN, LOGOUT, BOOKING_CREATED, ROUTE_UPDATED, USER_DELETED.

Sr.	Name	Data Type	Constraint	Description
03	Affected_Entity	VARCHAR(50)	NOT NULL	Name of the table/entity that was affected (e.g., TRIP, USER).
04	Affected_Record_ID	INT	OPTIONAL	Primary key of the affected record.
05	Old_Value	TEXT	OPTIONAL	record's state before the action.
06	New_Value	TEXT	OPTIONAL	record's state after the action.
07	IP_Address	VARCHAR(45)	OPTIONAL	IP address from which the action was performed.
08	Logged_At	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp when the action was recorded.

2.2.15.Fuel Consumption

Sr.	Name	Data Type	Constraint	Description
01	fuel_log_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for each fuel log entry.
02	vehicle_id (FK)	INT	NOT NULL	References the BUS that was refuelled.
03	liters	DECIMAL(6,2)	NOT NULL, > 0	Fuel quantity in litres.
04	cost	DECIMAL(10,2)	NOT NULL, ≥ 0	Total fuel cost in PKR.
05	filled_at	DATETIME	NOT NULL,	Timestamp when fuel was

			DEFAULT NOW()	filled.
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2.2.16. Vehicle Maintenance

Sr.	Name	Data Type	Constraint	Description
01	maintenance_id (PK)	INT	NOT NULL, AUTO_INCREMENT	Unique identifier for maintenance record.
02	vehicle_id (FK)	INT	NOT NULL	References the BUS.
03	maintenance_type	VARCHAR(80)	NOT NULL	Type (Oil Change, Tyre Replacement, etc.).
04	description	TEXT	OPTIONAL	Details of maintenance work.
05	maintenance_date	DATE	NOT NULL	Date maintenance was done.
06	maintenance_cost	DECIMAL(10,2)	NOT NULL, ≥ 0	Cost in PKR.

2.2. BUSINESS RULES:

The following business rules define constraints and operational policies governing the NUTMS database. These rules ensure data integrity, enforce access control, and reflect real-world transport management policies at Namal University.

1. User and Role Rules

- Every person using NUTMS must be registered as exactly one USER with one of four roles: Student, Faculty, Driver, or Administrator.
- A USER must belong to exactly one subtype — STUDENT, FACULTY, DRIVER, or ADMINISTRATOR. A user cannot belong to more than one subtype simultaneously (disjoint specialization).
- A STUDENT account is created by an ADMINISTRATOR; self-registration is not permitted.
- A USER account can be deactivated by an ADMINISTRATOR, but records must be retained for at least one academic year for audit purposes.
- A DRIVER must have a valid, non-expired driving license to be assigned to any SCHEDULE.

2. Vehicle and Route Rules

- A **Vehicle** must have a seating capacity greater than zero and a unique registration plate.
- A ROUTE must contain at least two STOPS: a starting stop and a terminal stop.
- Each STOP belongs to exactly one ROUTE and has a unique sequence number within that route.
- A **Vehicle** marked 'Under Maintenance' or 'Retired' cannot be assigned to any new SCHEDULE.
- A ROUTE marked inactive cannot have new SCHEDULEs created for it.

3. Schedule and Trip Rules

- A SCHEDULE must be associated with exactly one ROUTE and exactly one BUS.
- A BUS cannot have two overlapping SCHEDULEs on the same day and time.
- A DRIVER can be assigned to at most one active TRIP at any given time.
- A TRIP is generated from a SCHEDULE and must reference the same BUS and DRIVER as the SCHEDULE from which it originates.
- A TRIP status must progress in order: Scheduled → In Progress → Completed (or Cancelled); backward transitions are not permitted.
- A TRIP cannot be set to 'Completed' without an actual arrival time being recorded.

4. GPS Tracking Rules

- A GPS_LOG entry must be associated with exactly one BUS and one active TRIP.
- GPS location updates must be recorded at least every 30 seconds during an active TRIP.
- GPS data older than 90 days may be archived but must not be deleted within one academic year.

5. Special Vehicle Booking Rules

- Only FACULTY members may submit special vehicle BOOKING requests; STUDENTS may not.
- A FACULTY member may not have more than three PENDING bookings at any given time.
- A BOOKING must be submitted at least 24 hours before the requested trip date.
- A BOOKING can only be approved or rejected by an ADMINISTRATOR.
- An approved BOOKING must have an assigned BUS and DRIVER before the trip date.
- A BOOKING can only be cancelled by the requesting FACULTY member if it is still in Pending status.

6. Notification Rules

- A NOTIFICATION can be targeted to one or more specific USERS, or broadcast to all registered users.
- Emergency notifications must be sent to all active users regardless of individual preferences.
- A booking status change (approval or rejection) must automatically generate a NOTIFICATION to the requesting FACULTY member.
- A schedule delay exceeding 10 minutes must automatically trigger a delay NOTIFICATION to all users affected by that SCHEDULE.

7. Audit Rules

- Every login attempt (successful or failed) must be recorded in the AUDIT_LOG.
- Any change to a USER, ROUTE, SCHEDULE, BUS, or BOOKING record must generate an AUDIT_LOG entry capturing the before and after state.
- AUDIT_LOG records are immutable — they cannot be updated or deleted by any user role, including ADMINISTRATOR.
- AUDIT_LOG data must be retained for a minimum of one academic year.

2.3. ENTITY RELATIONSHIP DIAGRAM:

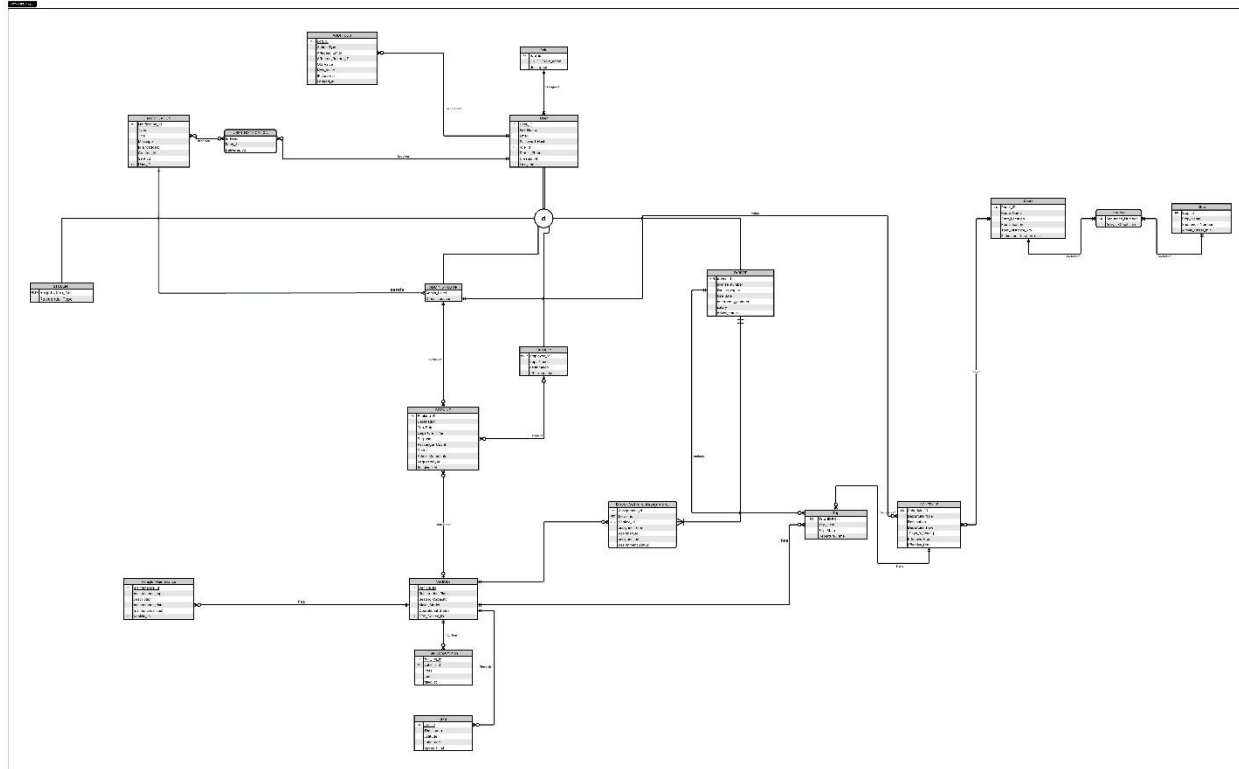


Figure 1: EERD of the Namal University Transport Management System (NUTMS)

ERD Link :

https://lucid.app/lucidchart/56b6e693-6a6c-4d78-89f5-a75e7660e4f5/edit?viewport_loc=11125%2C-1865%2C3276%2C1526%2C0_0&invitationId=inv_346d5c52-ab62-4ada-bd95-16d1ab9e4c9f

APPENDIX: AI USAGE AND PROMPTS

The following table documents all AI tools used during the development of this report, including the prompts used and their purpose.

Tool	Purpose	Prompt / Usage Description	Access Date
Claude	Document structure, entity/attribute identification, business rules, ERD prompt generation	Uploaded the SRS, DB Proposal, Milestone 2 instructions, ERD PPT slides, and EERD PPT slides. Asked to make the complete database design report according to the given template and generate a comprehensive Lucid AI ERD prompt.	May 2026
ChatGPT	Understanding ERD/EERD notation rules and Chen notation conventions	Explain Chen notation for ERDs: how to represent strong entities, weak entities, multi-valued attributes, derived attributes, composite attributes, relationships, participation constraints, and supertype/subtype hierarchies.	May 2026
ChatGPT	Reviewing data dictionary entries for completeness	Given the following list of entities for a university transport management system, suggest appropriate attributes, data types, and constraints for each entity following database design best practices.	May 2026